1) a)

$$|\mathbf{A}| = \begin{vmatrix} 4 & 40 \\ 12 & 600 \end{vmatrix}$$
$$= 4 \times 600 - 40 \times 12$$
$$= 2400 - 480$$
$$= 1920$$

1) b)

$$10\begin{vmatrix} 4 & 4 \\ 12 & 60 \end{vmatrix}$$
$$= 10(4 \times 60 - 4 \times 12)$$
$$= 10(240 - 48)$$
$$= 10 \times 192$$
$$= 1920$$

1) c)

$$40\begin{vmatrix} 1 & 1 \\ 12 & 60 \end{vmatrix}$$
$$= 40(1 \times 60 - 1 \times 12)$$
$$= 40(60 - 12)$$
$$= 40 \times 48$$
$$= 1920$$

1) d)

$$480\begin{vmatrix} 1 & 1 \\ 1 & 5 \end{vmatrix}$$
$$= 480(1 \times 5 - 1 \times 1)$$
$$= 480(5 - 1)$$
$$= 480 \times 4$$
$$= 1920$$

2) a)

$$|\mathbf{A}| = \begin{vmatrix} 120 & 21 \\ 150 & 35 \end{vmatrix}$$
$$= 120 \times 35 - 21 \times 150$$
$$= 4200 - 3150$$
$$= 1050$$

2) b)

$$2\begin{vmatrix}60 & 21\\75 & 35\end{vmatrix}$$
$$= 2(60 \times 35 - 21 \times 75)$$
$$= 2(2100 - 1575)$$
$$= 2 \times 525$$
$$= 1050$$

2) c)

$$6\begin{vmatrix}20 & 21\\25 & 35\end{vmatrix}$$
$$= 6(20 \times 35 - 21 \times 25)$$
$$= 6(700 - 525)$$
$$= 6 \times 175$$
$$= 1050$$

2) d)

$$30\begin{vmatrix}4 & 21\\5 & 35\end{vmatrix}$$
$$= 30(4 \times 35 - 21 \times 5)$$
$$= 30(140 - 125)$$
$$= 30 \times 15$$
$$= 1050$$

2) e)

$$150\begin{vmatrix}4 & 21\\1 & 7\end{vmatrix}$$
$$= 150(4 \times 7 - 21 \times 1)$$
$$= 150(28 - 21)$$
$$= 150 \times 7$$
$$= 1050$$

2) f)

$$1050\begin{vmatrix}4 & 3\\ 1 & 1\end{vmatrix}$$
$$= 1050(4 \times 1 - 3 \times 1)$$
$$= 1050(4 - 3)$$
$$= 1050 \times 1$$
$$= 1050$$

2) g)

$$|\mathbf{A}| = \begin{vmatrix}120 & 21\\ 150 & 35\end{vmatrix}$$
$$= 2\begin{vmatrix}60 & 21\\ 75 & 35\end{vmatrix}$$
$$= 2 \times 3\begin{vmatrix}20 & 21\\ 25 & 35\end{vmatrix}$$
$$= 2 \times 3 \times 5\begin{vmatrix}4 & 21\\ 5 & 35\end{vmatrix}$$
$$= 2 \times 3 \times 5 \times 5\begin{vmatrix}4 & 21\\ 1 & 7\end{vmatrix}$$
$$= 2 \times 3 \times 5 \times 5 \times 7\begin{vmatrix}4 & 3\\ 1 & 1\end{vmatrix}$$
$$= 2 \times 3 \times 5 \times 5 \times 7 \times 1$$
$$\therefore |\mathbf{A}| = 1050 = 2 \times 3 \times 5^2 \times 7$$

3) a)

$$|\mathbf{A}| = \begin{vmatrix}k & k+1\\ 2k & 4(k+1)\end{vmatrix}$$
$$= k \times 4(k+1) - (k+1) \times 2k$$
$$= 4k^2 + 4k - 2k^2 - 2k$$
$$= 2k^2 + 2k$$
$$= 2k(k+1)$$

3) b)

$$k\begin{vmatrix}1 & k+1\\ 2 & 4(k+1)\end{vmatrix}$$
$$= k(1 \times 4(k+1) - (k+1) \times 2)$$
$$= k(4k + 4 - 2k - 2)$$
$$= k(2k + 2)$$
$$= 2k(k+1)$$

3) c)

$$2k\begin{vmatrix}1 & k+1\\ 1 & 2(k+1)\end{vmatrix}$$
$$= 2k(1 \times 2(k+1) - (k+1) \times 1)$$
$$= 2k(2k+2-k-1)$$
$$= 2k(k+1)$$

3) d)

$$2k(k+1)\begin{vmatrix}1 & 1\\ 1 & 2\end{vmatrix}$$
$$= 2k(k+1)(1 \times 2 - 1 \times 1)$$
$$= 2k(k+1) \times 1$$
$$= 2k(k+1)$$

3) e)

$$|\mathbf{A}| = 2k(k+1)$$

4) a)

$$|\mathbf{A}| = \begin{vmatrix}18k-9 & 3-k\\ 6k-3 & 2k-6\end{vmatrix}$$
$$= (18k-9) \times (2k-6) - (3-k) \times (6k-3)$$
$$= 36k^2 - 108k - 18k + 54 - 18k + 9 + 6k^2 - 3k$$
$$= 42k^2 - 147k + 63$$
$$= 21(2k-1)(k-3)$$

4) b)

$$(2k-1)\begin{vmatrix}9 & 3-k\\ 3 & 2k-6\end{vmatrix}$$
$$= (2k-1)(9 \times (2k-6) - (3-k) \times 3)$$
$$= (2k-1)(18k-54-9+3k)$$
$$= (2k-1)(21k-63)$$
$$= 21(2k-1)(k-3)$$

4) c)

$3(2k-1)\begin{vmatrix}3 & 3-k\\ 1 & 2k-6\end{vmatrix}$

$= 3(2k-1)(3\times(2k-6)-(3-k)\times 1)$

$= 3(2k-1)(6k-18-3+k)$

$= 3(2k-1)(7k-21)$

$= 21(2k-1)(k-3)$

4) d)

$3(2k-1)(3-k)\begin{vmatrix}3 & 1\\ 1 & -2\end{vmatrix}$

$= 3(2k-1)(3-k)(3\times(-2)-1\times 1)$

$= 3(2k-1)(3-k)\times(-7)$

$= -21(2k-1)(3-k)$

$= 21(2k-1)(k-3)$

5)

$|\mathbf{A}| = \begin{vmatrix}4k & 6k-2\\ 20k & 9k-3\end{vmatrix}$

Factor $4k$ out of C1:

$= 4k\begin{vmatrix}1 & 2(3k-1)\\ 5 & 3(3k-1)\end{vmatrix}$

Factor $(3k-1)$ out of C2:

$= 4k(3k-1)\begin{vmatrix}1 & 2\\ 5 & 3\end{vmatrix}$

Then:

$= 4k(3k-1)(1\times 3-2\times 5)$

$= 4k(3k-1)(3-10)$

$= 4k(3k-1)\times -7$

$= -28k(3k-1)$

6)

$|\mathbf{A}| = 0 \Rightarrow \begin{vmatrix} 4-8k & 5-10k \\ 3k-6 & 4k-8 \end{vmatrix} = 0$

$\Rightarrow \begin{vmatrix} 4(1-2k) & 5(1-2k) \\ 3(k-2) & 4(k-2) \end{vmatrix} = 0$

$\Rightarrow (1-2k)\begin{vmatrix} 4 & 5 \\ 3(k-2) & 4(k-2) \end{vmatrix} = 0$

$\Rightarrow (1-2k)(k-2)\begin{vmatrix} 4 & 5 \\ 3 & 4 \end{vmatrix} = 0$

$\Rightarrow (1-2k)(k-2)(4\times 4 - 5\times 3) = 0$

$\Rightarrow (1-2k)(k-2) = 0$

$\Rightarrow k = \frac{1}{2}$ or $k = 2$

7) a)

$|\mathbf{A}| = \begin{vmatrix} 10 & 8 \\ 8 & 8 \end{vmatrix} = 10\times 8 - 8\times 8 = 80 - 64 = 16$

7) b)

$|\mathbf{A}| = \begin{vmatrix} 10 & 8 \\ 8 & 8 \end{vmatrix}$

Factor 2 out of R1:

$|\mathbf{A}| = 2\begin{vmatrix} 5 & 4 \\ 8 & 8 \end{vmatrix}$

Factor 8 out of R2:

$|\mathbf{A}| = 16\begin{vmatrix} 5 & 4 \\ 1 & 1 \end{vmatrix}$

Let the new $\mathrm{R1} = \mathrm{R1} - 4\mathrm{R2}$:

$|\mathbf{A}| = 16\begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix}$

7) c)

$|\mathbf{A}| = 16\begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} = 16\times(1\times 1 - 0\times 1) = 16\times 1 = 16$

8) a)

$|\mathbf{A}| = \begin{vmatrix} 16 & 20 \\ 9 & 5 \end{vmatrix} = 16\times 5 - 20\times 9 = 80 - 180 = -100$

8) b)

$|\mathbf{A}| = \begin{vmatrix} 16 & 20 \\ 9 & 5 \end{vmatrix}$

Factor 5 out of C2:

$|\mathbf{A}| = 5\begin{vmatrix} 16 & 4 \\ 9 & 1 \end{vmatrix}$

Let the new $\mathrm{C1} = \mathrm{C1} - 4\mathrm{C2}$:

$|\mathbf{A}| = 5\begin{vmatrix} 0 & 4 \\ 5 & 1 \end{vmatrix}$

8) c)

$|\mathbf{A}| = 5\begin{vmatrix} 0 & 4 \\ 5 & 1 \end{vmatrix} = 5 \times (0 \times 1 - 4 \times 5) = 5 \times (-20) = -100$

9) a)

$|\mathbf{A}| = \begin{vmatrix} 40 & 35 \\ 55 & 25 \end{vmatrix} = 40 \times 25 - 35 \times 55 = 1000 - 1925 = -925$

9) b)

$|\mathbf{A}| = \begin{vmatrix} 40 & 35 \\ 55 & 25 \end{vmatrix}$

Let the new $\mathrm{C1} = \mathrm{C1} - 2\mathrm{C2}$:

$|\mathbf{A}| = \begin{vmatrix} -30 & 35 \\ 5 & 25 \end{vmatrix}$

Let the new $\mathrm{R1} = \mathrm{R1} + 6\mathrm{R2}$:

$|\mathbf{A}| = \begin{vmatrix} 0 & 185 \\ 5 & 25 \end{vmatrix}$

9) c)

$|\mathbf{A}| = \begin{vmatrix} 0 & 185 \\ 5 & 25 \end{vmatrix} = 0 \times 25 - 185 \times 5 = -925$

10) a)

$|\mathbf{A}| = \begin{vmatrix} 4k+5 & 53 \\ k & 12 \end{vmatrix} = (4k+5) \times 12 - 53 \times k = 48k + 60 - 53k = 60 - 5k$

10) b)

$|\mathbf{A}| = \begin{vmatrix} 4k+5 & 53 \\ k & 12 \end{vmatrix}$

Let the new $\mathrm{R1} = \mathrm{R1} - 4\mathrm{R2}$:

$|\mathbf{A}| = \begin{vmatrix} 5 & 5 \\ k & 12 \end{vmatrix}$

Let the new $\mathrm{C2} = \mathrm{C2} - \mathrm{C1}$:

$|\mathbf{A}| = \begin{vmatrix} 5 & 0 \\ k & 12-k \end{vmatrix}$

10) c)

$|\mathbf{A}| = \begin{vmatrix} 5 & 0 \\ k & 12-k \end{vmatrix} = 5 \times (12-k) - 0 \times k = 60 - 5k$

11) a)

Grace has made a mistake at step (3), where she has attempted to factor 3 out of column 1. 1 is the top-left element in the matrix by stage (2), so if 3 were factored out it would become $\frac{1}{3}$ rather than staying as 1.

11) b)

$|\mathbf{A}| = \begin{vmatrix} q+1 & -5p \\ 3q+3 & 15p \end{vmatrix} = \begin{vmatrix} q+1 & -5p \\ 3(q+1) & 15p \end{vmatrix}$

Factor $(q+1)$ out of $\mathrm{C1}$:

$= (q+1)\begin{vmatrix} 1 & -5p \\ 3 & 15p \end{vmatrix}$

Factor $5p$ out of $\mathrm{C2}$:

$= 5p(q+1)\begin{vmatrix} 1 & -1 \\ 3 & 3 \end{vmatrix}$

Let the new $\mathrm{C1} = \mathrm{C1} - \mathrm{C2}$:

$= 5p(q+1)\begin{vmatrix} 2 & -1 \\ 0 & 3 \end{vmatrix}$

Then:

$= 30p(q+1)$

12) a)

$$\begin{vmatrix} x & y \\ x^2 & y^2 \end{vmatrix}$$

Factor x out of C1:

$$= x\begin{vmatrix} 1 & y \\ x & y^2 \end{vmatrix}$$

Factor y out of C2:

$$= xy\begin{vmatrix} 1 & 1 \\ x & y \end{vmatrix}$$

Let the new $\mathrm{C2} = \mathrm{C2} - \mathrm{C1}$:

$$= xy\begin{vmatrix} 1 & 0 \\ x & y - x \end{vmatrix}$$

Then:

$$= xy(y - x)$$

12) b)

$$\begin{vmatrix} x^2 & y^2 \\ xy & xy \end{vmatrix}$$

Factor x out of C1:

$$= x\begin{vmatrix} x & y^2 \\ y & xy \end{vmatrix}$$

Factor y out of C2:

$$= xy\begin{vmatrix} x & y \\ y & x \end{vmatrix}$$

Let the new $\mathrm{R1} = \mathrm{R1} + \mathrm{R2}$:

$$= xy\begin{vmatrix} x + y & x + y \\ y & x \end{vmatrix}$$

Factor $(x + y)$ out of R1:

$$= xy(x + y)\begin{vmatrix} 1 & 1 \\ y & x \end{vmatrix}$$

Let the new $\mathrm{C2} = \mathrm{C2} - \mathrm{C1}$:

$$= xy(x + y)\begin{vmatrix} 1 & 0 \\ y & x - y \end{vmatrix}$$

Then:

$$= xy(x + y)(x - y)$$

12) c)

$$\begin{vmatrix} xy & xy^2 \\ x^2 & xy \end{vmatrix}$$

Factor x out of $\mathrm{C1}$:

$$= x \begin{vmatrix} y & xy^2 \\ x & xy \end{vmatrix}$$

Factor y out of $\mathrm{C2}$:

$$= xy \begin{vmatrix} y & xy \\ x & x \end{vmatrix}$$

Factor y out of $\mathrm{R1}$:

$$= xy^2 \begin{vmatrix} 1 & x \\ x & x \end{vmatrix}$$

Factor x out of $\mathrm{R2}$:

$$= x^2y^2 \begin{vmatrix} 1 & x \\ 1 & 1 \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} - \mathrm{R1}$:

$$= x^2y^2 \begin{vmatrix} 1 & x \\ 0 & 1-x \end{vmatrix}$$

Then:

$$= x^2y^2(1-x)$$

12) d)

$$\begin{vmatrix} yz & xy \\ xyz & xyz \end{vmatrix}$$

Factor y out of $\mathrm{R1}$:

$$= y \begin{vmatrix} z & x \\ xyz & xyz \end{vmatrix}$$

Factor xyz out of $\mathrm{R2}$:

$$= xy^2z \begin{vmatrix} z & x \\ 1 & 1 \end{vmatrix}$$

Let the new $\mathrm{C1} = \mathrm{C1} - \mathrm{C2}$:

$$= xy^2z \begin{vmatrix} z-x & x \\ 0 & 1 \end{vmatrix}$$

Then:

$$= xy^2z(z-x)$$

12) e)

$$\begin{vmatrix} xyz & yz \\ xy & xyz \end{vmatrix}$$

Factor y out of R1:

$$= y \begin{vmatrix} xz & z \\ xy & xyz \end{vmatrix}$$

Factor xy out of R2:

$$= xy^2 \begin{vmatrix} xz & z \\ 1 & z \end{vmatrix}$$

Let the new $\mathrm{R1} = \mathrm{R1} - \mathrm{R2}$:

$$= xy^2 \begin{vmatrix} xz - 1 & 0 \\ 1 & z \end{vmatrix}$$

Then:

$$= xy^2z(xz - 1)$$

12) f)

$$\begin{vmatrix} x^2y & xz^2 \\ xy^2 & xyz^2 \end{vmatrix}$$

Factor z^2 out of C2:

$$= z^2 \begin{vmatrix} x^2y & x \\ xy^2 & xy \end{vmatrix}$$

Factor xy out of R2:

$$= xyz^2 \begin{vmatrix} x^2y & x \\ y & 1 \end{vmatrix}$$

Factor y out of C1:

$$= xy^2z^2 \begin{vmatrix} x^2 & x \\ 1 & 1 \end{vmatrix}$$

Factor x out of R1:

$$= x^2y^2z^2 \begin{vmatrix} x & 1 \\ 1 & 1 \end{vmatrix}$$

Let the new $\mathrm{R1} = \mathrm{R1} - \mathrm{R2}$:

$$= x^2y^2z^2 \begin{vmatrix} x - 1 & 0 \\ 1 & 1 \end{vmatrix}$$

Then:

$$= x^2y^2z^2(x - 1)$$

12) g)

$$\begin{vmatrix} x^2y^2z & xy^2z^2 \\ x^2yz^2 & xyz \end{vmatrix}$$

Factor xyz out of R1:

$$= xyz \begin{vmatrix} xy & yz \\ x^2yz^2 & xyz \end{vmatrix}$$

Factor xyz out of R2:

$$= x^2y^2z^2 \begin{vmatrix} xy & yz \\ xz & 1 \end{vmatrix}$$

Factor x out of C1:

$$= x^3y^2z^2 \begin{vmatrix} y & yz \\ z & 1 \end{vmatrix}$$

Factor y out of R1:

$$= x^3y^3z^2 \begin{vmatrix} 1 & z \\ z & 1 \end{vmatrix}$$

Let the new $\mathrm{R1} = \mathrm{R1} + \mathrm{R2}$:

$$= x^3y^3z^2 \begin{vmatrix} 1+z & 1+z \\ z & 1 \end{vmatrix}$$

Factor $(1+z)$ out of R1:

$$= x^3y^3z^2(1+z) \begin{vmatrix} 1 & 1 \\ z & 1 \end{vmatrix}$$

Let the new $\mathrm{R1} = \mathrm{R1} - \mathrm{R2}$:

$$= x^3y^3z^2(1+z) \begin{vmatrix} 1-z & 0 \\ z & 1 \end{vmatrix}$$

Then:

$$= x^3y^3z^2(1+z)(1-z)$$

13) a)

$$\begin{vmatrix} p & -p-2q \\ 4p+5q & p+2q \end{vmatrix} = \begin{vmatrix} p & -(p+2q) \\ 4p+5q & p+2q \end{vmatrix}$$

Factor $(p+2q)$ out of C1:

$$= (p+2q)\begin{vmatrix} p & -1 \\ 4p+5q & 1 \end{vmatrix}$$

Let the new $\mathrm{R1} = \mathrm{R1} + \mathrm{R2}$:

$$= (p+2q)\begin{vmatrix} 5p+5q & 0 \\ 4p+5q & 1 \end{vmatrix} = (p+2q)\begin{vmatrix} 5(p+q) & 0 \\ 4(p+q) & 1 \end{vmatrix}$$

Factor $(p+q)$ out of C1:

$$= (p+2q)(p+q)\begin{vmatrix} 5 & 0 \\ 4 & 1 \end{vmatrix}$$

Then:

$$= 5(p+2q)(p+q)$$

13) b)

$$\begin{vmatrix} 2q-2p & 3p-3q \\ 14q-6p & 2p \end{vmatrix} = \begin{vmatrix} -2(p-q) & 3(p-q) \\ 14q-6p & 2p \end{vmatrix}$$

Factor $(p-q)$ out of R1:

$$= (p-q)\begin{vmatrix} -2 & 3 \\ 14q-6p & 2p \end{vmatrix}$$

Factor 2 out of R2:

$$= 2(p-q)\begin{vmatrix} -2 & 3 \\ 7q-3p & p \end{vmatrix}$$

Let the new $\mathrm{C1} = \mathrm{C1} + 3\mathrm{C2}$:

$$= 2(p-q)\begin{vmatrix} 7 & 3 \\ 7q & p \end{vmatrix}$$

Factor 7 out of C1:

$$= 14(p-q)\begin{vmatrix} 1 & 3 \\ q & p \end{vmatrix}$$

Let the new $\mathrm{C2} = \mathrm{C2} - 3\mathrm{C1}$:

$$= 14(p-q)\begin{vmatrix} 1 & 0 \\ q & p-3q \end{vmatrix}$$

Then:

$$= 14(p-q)(p-3q)$$

13) c)

$$\begin{vmatrix} 16p+5q-13 & 4-8p \\ 6q-12p & 6p-3 \end{vmatrix} = \begin{vmatrix} 16p+5q-13 & -4(2p-1) \\ 6q-12p & 3(2p-1) \end{vmatrix}$$

Factor $(2p-1)$ out of C1:

$$= (2p-1)\begin{vmatrix} 16p+5q-13 & -4 \\ 6q-12p & 3 \end{vmatrix}$$

Let the new $\mathrm{R1} = \mathrm{R1} + \frac{4}{3} \times \mathrm{R2}$:

$$= (2p-1)\begin{vmatrix} 13q-13 & 0 \\ 6q-12p & 3 \end{vmatrix}$$

Factor $13(q-1)$ out of R1:

$$= 13(2p-1)(q-1)\begin{vmatrix} 1 & 0 \\ 6q-12p & 3 \end{vmatrix}$$

Then:

$$= 39(2p-1)(q-1)$$

13) d)

$$\begin{vmatrix} 3-2q & 4q-8p \\ 6-4q & 8q+24p-30 \end{vmatrix} = \begin{vmatrix} 3-2q & 4q-8p \\ 2(3-2q) & 8q+24p-30 \end{vmatrix}$$

Factor $(3-2q)$ out of C1:

$$= (3-2q)\begin{vmatrix} 1 & 4q-8p \\ 2 & 8q+24p-30 \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} - 2\mathrm{R1}$:

$$= (3-2q)\begin{vmatrix} 1 & 4q-8p \\ 0 & 40p-30 \end{vmatrix}$$

Factor $10(4p-3)$ out of R2:

$$= 10(3-2q)(4p-3)\begin{vmatrix} 1 & 4q-8p \\ 0 & 1 \end{vmatrix}$$

Then:

$$= 10(3-2q)(4p-3)$$

14)

$|\mathbf{A}| = xy\begin{vmatrix}1 & -3\\ 2 & 4\end{vmatrix}$

Multiply C1 by x:

$= y\begin{vmatrix}x & -3\\ 2x & 4\end{vmatrix}$

Multiply C2 by y:

$= \begin{vmatrix}x & -3y\\ 2x & 4y\end{vmatrix}$

Let the new $\mathrm{R2} = \mathrm{R2} - 2\mathrm{R1}$:

$= \begin{vmatrix}x & -3y\\ 0 & 10y\end{vmatrix}$ as required

15)

$|\mathbf{A}| = x(x-1)y\begin{vmatrix}2 & -3\\ 3 & -1\end{vmatrix}$

Multiply C1 by x:

$= (x-1)y\begin{vmatrix}2x & -3\\ 3x & -1\end{vmatrix}$

Multiply C2 by $(x-1)y$:

$= \begin{vmatrix}2x & -3(x-1)y\\ 3x & -(x-1)y\end{vmatrix} = \begin{vmatrix}2x & 3y-3xy\\ 3x & y-xy\end{vmatrix}$

Multiply R1 by 3:

$= \frac{1}{3}\begin{vmatrix}6x & 9y-9xy\\ 3x & y-xy\end{vmatrix}$

Multiply R2 by 2:

$= \frac{1}{6}\begin{vmatrix}6x & 9y-9xy\\ 6x & 2y-2xy\end{vmatrix}$

Factor 6 out of C1:

$= \begin{vmatrix}x & 9y-9xy\\ x & 2y-2xy\end{vmatrix}$

Let the new $\mathrm{R1} = \mathrm{R1} - \mathrm{R2}$:

$= \begin{vmatrix}0 & 7y-7xy\\ x & 2y-2xy\end{vmatrix}$

Let the new $\mathrm{C2} = \mathrm{C2} + \mathrm{C1}$:

$= \begin{vmatrix}0 & 7y-7xy\\ x & x+2y-2xy\end{vmatrix}$ as required

16)

$\mathbf{A}$ is singular $\Rightarrow |\mathbf{A}| = 0$

$$\begin{vmatrix} x-5a & -4a \\ 14x+2a & 9x+7a \end{vmatrix} = 0$$

Multiply C1 by 4:

$$\Rightarrow \frac{1}{4}\begin{vmatrix} 4x-20a & -4a \\ 56x+8a & 9x+7a \end{vmatrix} = 0$$

Let the new $\mathrm{C1} = \mathrm{C1} - 5\mathrm{C2}$:

$$\Rightarrow \frac{1}{4}\begin{vmatrix} 4x & -4a \\ 11x-27a & 9x+7a \end{vmatrix} = 0$$

Let the new $\mathrm{C2} = \mathrm{C2} + \mathrm{C1}$:

$$\Rightarrow \frac{1}{4}\begin{vmatrix} 4x & 4x-4a \\ 11x-27a & 20x-20a \end{vmatrix} = 0$$

$$\Rightarrow \frac{1}{4}\begin{vmatrix} 4x & 4(x-a) \\ 11x-27a & 20(x-a) \end{vmatrix} = 0$$

Factor $(x-a)$ out of C2:

$$\Rightarrow \frac{1}{4}(x-a)\begin{vmatrix} 4x & 4 \\ 11x-27a & 20 \end{vmatrix} = 0$$

Multiply R1 by 5:

$$\Rightarrow \frac{1}{20}(x-a)\begin{vmatrix} 20x & 20 \\ 11x-27a & 20 \end{vmatrix} = 0$$

Let the new $\mathrm{R1} = \mathrm{R1} - \mathrm{R2}$:

$$\Rightarrow \frac{1}{20}(x-a)\begin{vmatrix} 9x+27a & 0 \\ 11x-27a & 20 \end{vmatrix} = 0$$

Then:

$$\Rightarrow \frac{1}{20}(x-a) \times (9x+27a) \times 20 = 0$$

$$\Rightarrow 9(x-a)(x+3a) = 0$$

$$\Rightarrow (x-a)(x+3a) = 0$$

$\therefore x = a$ or $x = -3a$

17)

$\mathbf{A}$ is singular $\Rightarrow |\mathbf{A}| = 0$

$$\begin{vmatrix} 8a^2 - 2x^2 & 3x^2 - 6ax - 24a^2 \\ 5ax - 10a^2 & 8ax - 32a^2 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 2(4a^2 - x^2) & 3(x^2 - 2ax - 8a^2) \\ 5a(x - 2a) & 8a(x - 4a) \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} -2(x - 2a)(x + 2a) & 3(x - 4a)(x + 2a) \\ 5a(x - 2a) & 8a(x - 4a) \end{vmatrix} = 0$$

Factor a out of R2:

$$\Rightarrow a\begin{vmatrix} -2(x - 2a)(x + 2a) & 3(x - 4a)(x + 2a) \\ 5(x - 2a) & 8(x - 4a) \end{vmatrix} = 0$$

Divide through by a:

$$\Rightarrow \begin{vmatrix} -2(x - 2a)(x + 2a) & 3(x - 4a)(x + 2a) \\ 5(x - 2a) & 8(x - 4a) \end{vmatrix} = 0$$

Factor $(x - 2a)$ out of C1:

$$\Rightarrow (x - 2a)\begin{vmatrix} -2(x + 2a) & 3(x - 4a)(x + 2a) \\ 5 & 8(x - 4a) \end{vmatrix} = 0$$

Factor $(x + 2a)$ out of R1:

$$\Rightarrow (x + 2a)(x - 2a)\begin{vmatrix} -2 & 3(x - 4a) \\ 5 & 8(x - 4a) \end{vmatrix} = 0$$

Factor $(x - 4a)$ out of C2:

$$\Rightarrow (x + 2a)(x - 2a)(x - 4a)\begin{vmatrix} -2 & 3 \\ 5 & 8 \end{vmatrix} = 0$$

Divide through by $\begin{vmatrix} -2 & 3 \\ 5 & 8 \end{vmatrix}$ (as it is just a constant):

$$\Rightarrow (x + 2a)(x - 2a)(x - 4a) = 0$$

$\therefore x = -2a$ or $x = 2a$ or $x = 4a$

18)

$$\begin{vmatrix} 3a-4b-3c & a-2b-2c \\ -a+4b-c & a+2b-2c \end{vmatrix}$$

Let the new $\mathrm{R1} = \mathrm{R1} + \mathrm{R2}$:

$$= \begin{vmatrix} 2a-4c & 2a-4c \\ -a+4b-c & a+2b-2c \end{vmatrix}$$

Factor $(2a-4c)$ out of R1:

$$= (2a-4c)\begin{vmatrix} 1 & 1 \\ -a+4b-c & a+2b-2c \end{vmatrix}$$

Let the new $\mathrm{C2} = \mathrm{C2} - \mathrm{C1}$:

$$= (2a-4c)\begin{vmatrix} 1 & 0 \\ -a+4b-c & 2a-2b-c \end{vmatrix}$$

Then:

$$= (2a-4c)(2a-2b-c)$$

$$= 2(a-2c)(2a-2b-c)$$

19)

$$\begin{vmatrix} pq-4p+5q-20 & 3pq+15q \\ 3pq-12p-6q+24 & 4pq-10p-8q+20 \end{vmatrix}$$

$$= \begin{vmatrix} (p+5)(q-4) & 3q(p+5) \\ 3(pq-4p-2q+8) & 4pq-10p-8q+20 \end{vmatrix}$$

$$= \begin{vmatrix} (p+5)(q-4) & 3q(p+5) \\ 3(p-2)(q-4) & 4pq-10p-8q+20 \end{vmatrix}$$

Factor $(q-4)$ out of C1:

$$= (q-4)\begin{vmatrix} p+5 & 3q(p+5) \\ 3(p-2) & 4pq-10p-8q+20 \end{vmatrix}$$

Factor $(p+5)$ out of R1:

$$= (p+5)(q-4)\begin{vmatrix} 1 & 3q \\ 3p-6 & 4pq-10p-8q+20 \end{vmatrix}$$

Let the new $C2 = C2 + 6C1$:

$$= (p+5)(q-4)\begin{vmatrix} 1 & 3q+6 \\ 3p-6 & 4pq+8p-8q-16 \end{vmatrix}$$

$$= (p+5)(q-4)\begin{vmatrix} 1 & 3(q+2) \\ 3(p-2) & 4(pq+2p-2q-4) \end{vmatrix}$$

$$= (p+5)(q-4)\begin{vmatrix} 1 & 3(q+2) \\ 3(p-2) & 4(p-2)(q+2) \end{vmatrix}$$

Factor $(p-2)$ out of R2:

$$= (p-2)(p+5)(q-4)\begin{vmatrix} 1 & 3(q+2) \\ 3 & 4(q+2) \end{vmatrix}$$

Factor $(q+2)$ out of C2:

$$= (p-2)(p+5)(q+2)(q-4)\begin{vmatrix} 1 & 3 \\ 3 & 4 \end{vmatrix}$$

Let the new $R2 = R2 - 3R1$:

$$= (p-2)(p+5)(q+2)(q-4)\begin{vmatrix} 1 & 3 \\ 0 & -5 \end{vmatrix}$$

Then:

$$= -5(p-2)(p+5)(q+2)(q-4)$$

20) a)

$$\begin{vmatrix} x+3a+b & x+3a \\ 5a+b & 5a \end{vmatrix}$$

Let the new $\mathrm{C1} = \mathrm{C1} - \mathrm{C2}$:

$$= \begin{vmatrix} b & x+3a \\ b & 5a \end{vmatrix}$$

Factor b out of $\mathrm{C1}$:

$$= b\begin{vmatrix} 1 & x+3a \\ 1 & 5a \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} - \mathrm{R1}$:

$$= b\begin{vmatrix} 1 & x+3a \\ 0 & 2a-x \end{vmatrix}$$

Then:

$$= b(2a - x)$$

20) b)

$5a = 50 \Rightarrow a = 10$

$50 + b = 55 \Rightarrow b = 5$

$|\mathbf{A}| = \pm\frac{50}{20} = \pm\frac{5}{2}$

If $|\mathbf{A}| = \frac{5}{2}$:

$\Rightarrow 5(2 \times 10 - x) = \frac{5}{2}$

$\Rightarrow 20 - x = \frac{1}{2}$

$\Rightarrow x = \frac{39}{2}$

If $|\mathbf{A}| = -\frac{5}{2}$:

$\Rightarrow 5(2 \times 10 - x) = -\frac{5}{2}$

$\Rightarrow 20 - x = -\frac{1}{2}$

$\Rightarrow x = \frac{41}{2}$

$\therefore x = \frac{39}{2}$ or $x = \frac{41}{2}$